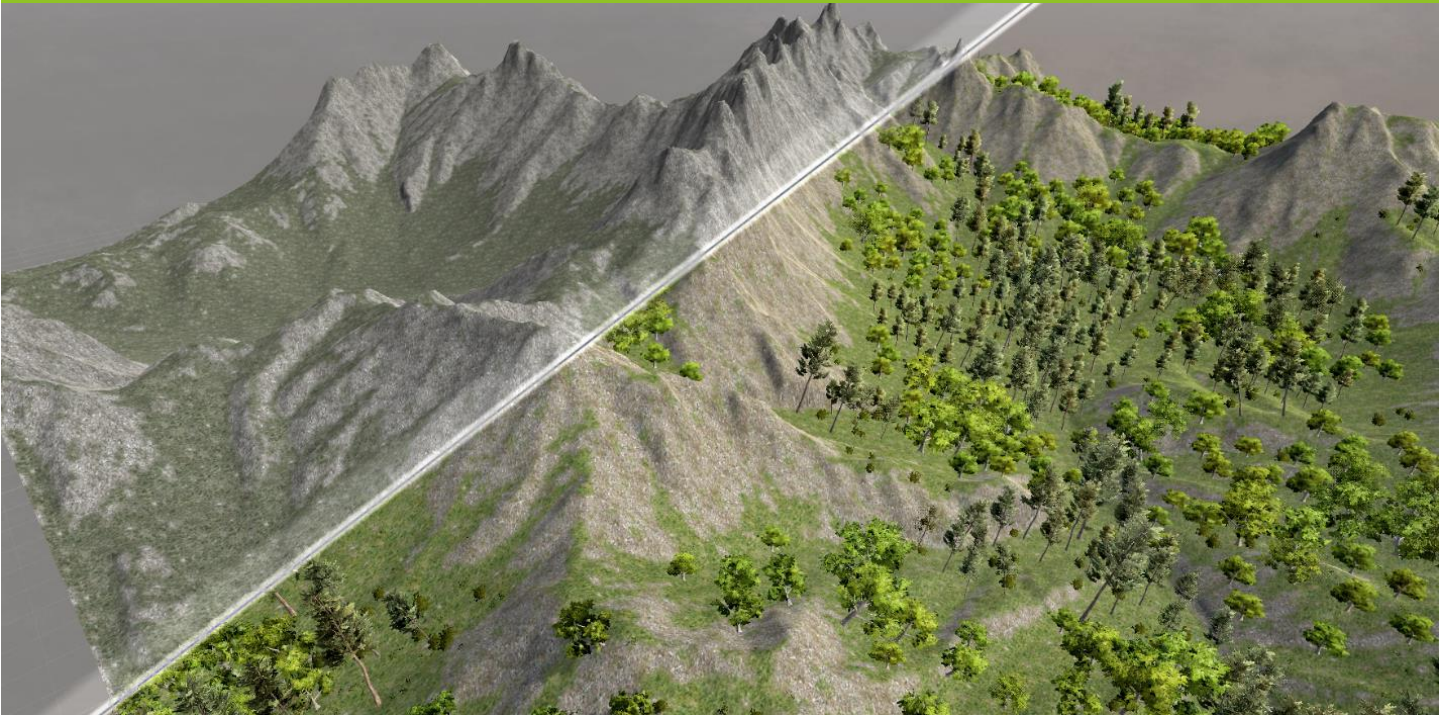


Landscaper 1.4



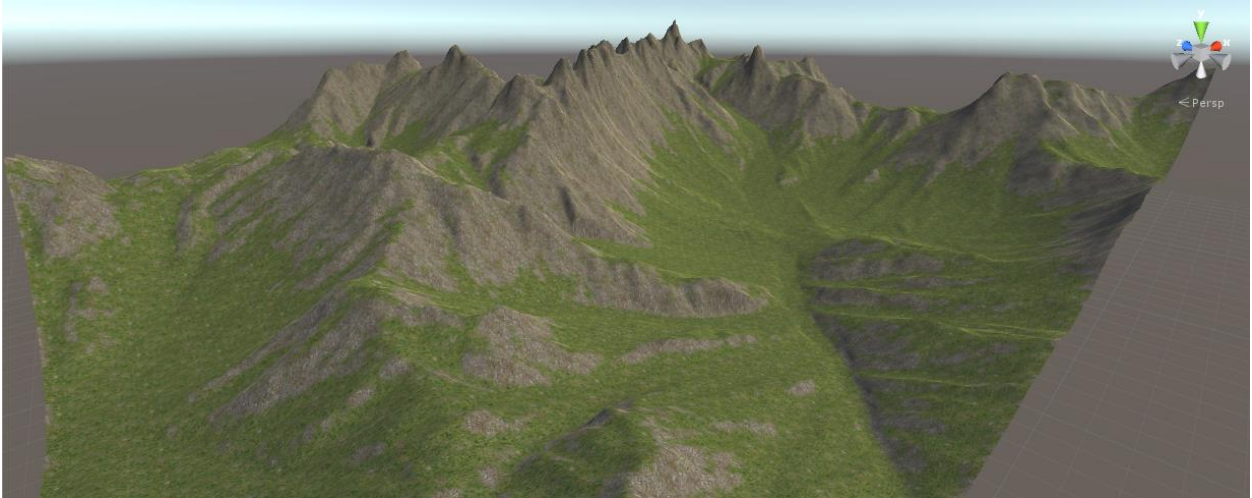
Unity Editor Extension by
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Quick Start

1. First, we'll need to make a terrain or other surface for Landscaper to plant our foliage on

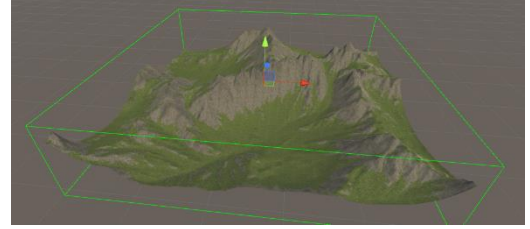


2. Next, we need a prefab that will represent a single instance of our foliage. For this example, we'll use one of the tree meshes from the standard assets included in Unity.
3. We then need to make a new "Species" asset which will provide all of the information for how our new foliage should be placed. In the menu, go to "Assets > Create > Landscaper > Species" to create a new species of foliage.



4. Our newly created species has a lot of parameters that can be tweaked which we'll go into in detail in a later section; for now though, there's only one field we're interested in. Press the "+" button to add a new archetype, and set the prefab field to the prefab we created in step 2.
5. There's one more asset we need to make before we can start placing foliage into the world; we need a Biome to describe which types of foliage can appear in an area. Go to "Assets > Create > Landscaper > Biome" to create one. In the settings for this new biome, press the "+" button to add a new species to the list, select the species asset we just created.

6. Now we need to add a Foliage Area to the scene, you can do this from the menu under “GameObject > Landscaper > Foliage Area”. The green bounding box represents the area in which foliage instances will be placed, by default, the new volume will encompass all geometry in the scene, but you can move/scale the GameObject to your liking. The larger the area, the more foliage instances will be generated and the longer the process will take, so you might want to start with a smaller area to begin with.
7. With our foliage area selected, we need to set the Biome field in the inspector to the Biome asset we created in step 5. Press “Simulate” and your scene should be filled with foliage!

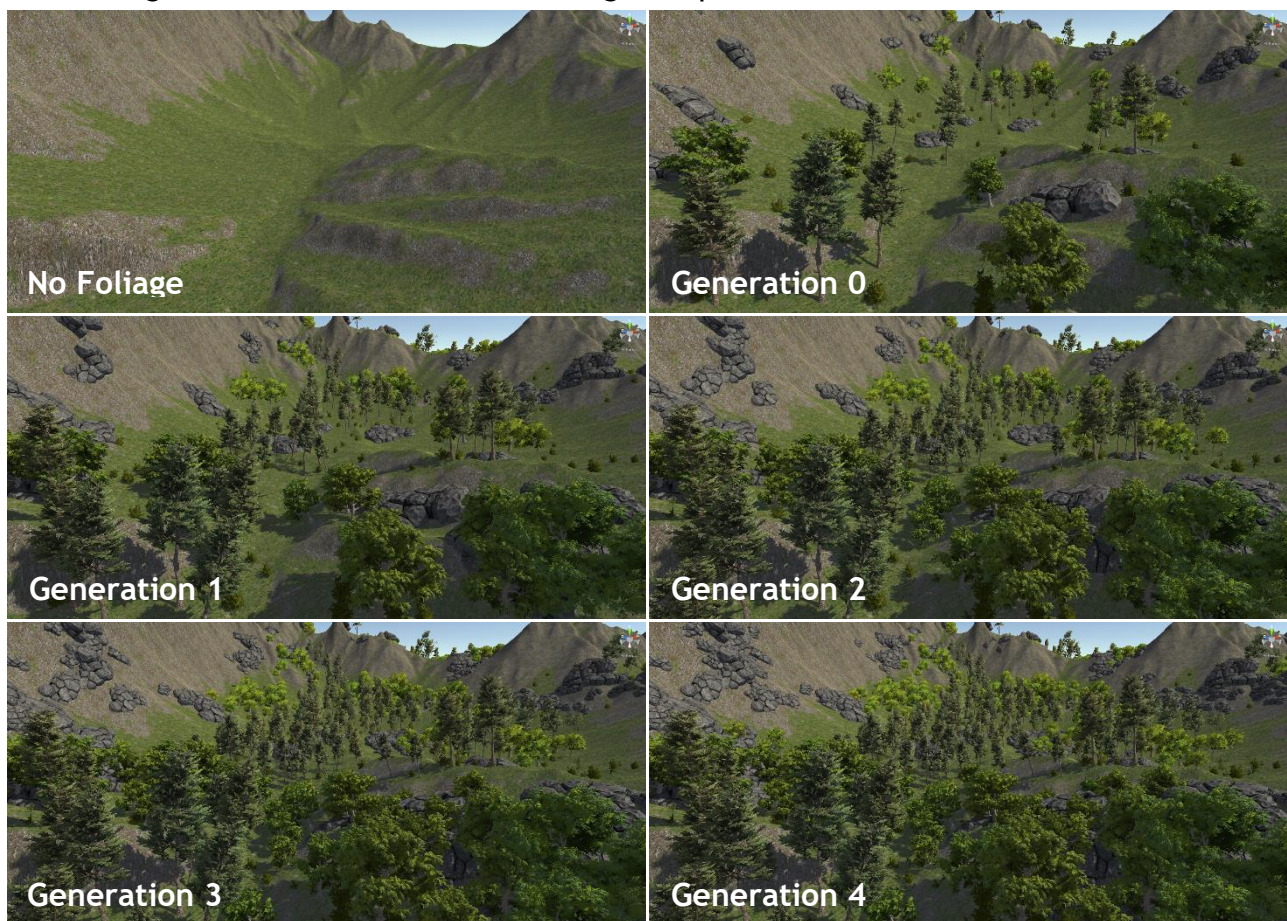


The Process

It's important to understand how Landscaper does what it does in order to make the best use of the settings available. Here's a brief overview of how the process works:

For each species, Landscaper plants a number of seeds at random in the foliage area. This number is based on the size of the volume and the “Initial Seed Density” parameter of the foliage species. Checks are performed to make sure the seed is in a valid place (again, according to the parameters in the species asset). This is the initial set or “generation 0”.

For each foliage instance placed in the previous step, a number of seeds are released which travel a certain distance before creating new foliage instances themselves. This process is repeated, and each new set of foliage created in this manner is considered a different generation. This allows for foliage to spread in a more natural fashion.



Note that Landscaper can be used to place other types of object such as the rocks in the above example.

Settings & Details

Species

A species in Landscaper acts as a single distinct “type” of foliage. The species controls which prefab is used to represent an instance of the foliage, where the instances can be placed, and how it interacts with other foliage in the scene.

Archetypes Archetypes are a combination of a prefab, and some settings to describe how Landscaper should use that prefab. An archetype makes up the physical/visual representation of a single foliage instance in the world. Each planted instance will pick an archetype from the list at random to represent itself.

Prefab	The prefab to place
Placement Mode	Defines how this archetype should be placed in the world. “GameObject” mode will always be used when an instance is placed on a non-terrain surface. <u>Terrain Tree</u> - More performant - Used for species that must have many instances placed <u>Game Object</u> - Can align to the surface normal - Can make full use of foliage created through the tree creator - Can contain scripts (could be used for interactable objects, for example)
Size by Generation	A range for the scale of an instance of this archetype. Younger (newer) generations will take their scale from the lower portion of the slider. This allows you to smoothly scale generations without having to specify generation overrides.
Colour Variation	Specifies how much a foliage instance can be darkened to add variation.

Align to Surface?	If true, foliage instances be rotated to be perpendicular to the surface they were planted on, otherwise, they will point directly upwards. Only works when using the “GameObject” placement method.
Initial Seed Density	How tightly-packed the initial set of foliage instances will be. The actual number of instances is also dependent on the volume of the bounding box for the foliage area used to spawn them.
Seed Travel Range	How far away a child instance can spawn from its parent.
Child Count (Min - Max)	How many children each instance of the foliage can have.
Collision Range	The collision radius for an instance of this species. No other foliage instances can be placed within this range.
Shade Range	The radius of the shadow that would be cast by an instance of this species. This is used to prevent other foliage instances from being planted within this radius (unless that particular species can grow in the shade)
Can Grow in Shade?	Can this species grow in the shade of another foliage instance?
Scale Variation (Min - Max)	The minimum and maximum scale multipliers that can be applied as random variation to each foliage instance, this is applied in addition to any generation-specific scale overrides.
Slope Angle (Min - Max)	The range of surface slope angles that an instance of this species can be planted on. Used to prevent instances from being placed on surfaces that are too steep, for example.
Altitude (Min - Max)	The range of altitudes this species can grow at. This is expressed as Y position in world-space.
Generations	A list of generation-specific settings. The first element in this list represents “generation 0” (or the initial set), each element after that represents the generation created by the previous. These settings are optional; if unspecified, the foliage instances will use default settings. You also don’t need an entry for every generation you will have - any generation beyond the number in this list will be clamped to use the last element. <i>E.g. With a list of 4 generations, the 5th generation will use the settings for generation 4.</i> Generations will not be used if their checkbox is not selected, this allows you to override generation 1, but not generation 0, for example.

Terrain Layers

We can control the density of foliage per terrain layer using the 'Terrain Layers' section of the species settings. If no layers are specified, the settings in this section will have no effect and foliage will grow everywhere. The default density is used for any layers not given a value explicitly.

Mode:

Allows us to choose how the layer mask is defined.

Terrain Layer (Default): Just use the terrain layer itself.

Texture: Use the albedo/diffuse texture.

Texture Index: Use the index of the terrain layer (0 for the first layer, 1 for the second, etc)

Combine Density

When we have two or more layers painted on an area where foliage is going to be planted, this option determines what density we use.

Assuming we have two layers: grass with a density of 1, and rock with a density of 0. For a potential spawn point that's 60% grass and 40% rock:

Blend (Default): Uses a blend of the two (or more) densities based on how much of each terrain layer is painted in the desired area.

$$\text{Density} = (0.6 \times 1) + (0.4 \times 0) = 0.6$$

Min: Uses the lowest density value. Density = 0

Use this if you don't want foliage to spawn if there's ANY rock.

Max: Uses the highest density value. Density = 1

Use this if you want foliage to spawn at full density if there's ANY grass.

Biome

A Biome in Landscaper acts as a way of grouping types of foliage that should inhabit an area. Each element in the list of species has two options: the species, and the weight. The species is just a Species asset as described above. The weight is a way to determine how many instances of that particular species of foliage are planted relative to the other species in the list. The value itself is not important (as long as it's greater than 0), what matters is how that value compares to the weights of the other species used by the biome.

Foliage Area

A Foliage Area is used to place instances of foliage from a chosen Biome into the scene. With all of the settings filled out, you can press the “Simulate” button in the inspector to generate instances of foliage in the world. The “Clear” button can be used to remove all of the created GameObjects.

Seed The initial seed passed to the random number generator. This allows you to re-generate the same batch of foliage twice; it also means that if we don't like how the foliage was generated the first time, we can change the seed and re-simulate for a different result (while still using the same settings for Species' and the Biome)

Generations Count	Indicates how many times the previous generation should spawn a new set of children. If set to 1, only the initial seeds will be planted. The larger this value, the more filled-out the area will become. Note that increasing the generations count exponentially increases the amount of foliage generated (thus, increasing the generation time). It's best to start with a fairly low number while testing & tweaking Species/Biome settings and then increase this value when generating the final set of foliage.
Bounds	The unscaled axis-aligned bounding box (AABB) that represents where foliage can be placed. This is translated and scaled according to the owning GameObject's Transform component
Foliage Layer	The layer to place new foliage instances on. While placing instances, Landscaper will use the “Ignore Raycast” layer for placing foliage GameObjects. Once finished, these will all be moved to the layer of your choosing.
Place Foliage On	Foliage will not be placed on any object not in any of these layers however raycasts may pass through and hit the terrain beneath, potentially causing foliage to be placed there.
Excluded Layers	Foliage will not be placed on an object in any of these layers and raycasts will be blocked, preventing foliage from being placed on anything below either.

Biome	The biome settings to use for this particular foliage area.
Max Tile Size	If the Foliage Area (green box) is larger than this size, it will be split into multiple smaller volumes (yellow boxes) to improve generation speed.
Density Scale	Controls the global density scale for foliage placed by this FoliageArea.
Limit Mode	Due to a restriction in Unity, any Terrain Tree instances placed after the first 65,535 (per-terrain) are not correctly handled in terms of LOD (level of detail) and will permanently show as a billboard. The Limit Mode defines how Landscaper should handle these situations. Note that this limit does not apply to foliage placed as GameObjects. - Ignore - <i>Landscaper will ignore the cap and continue to place terrain trees.</i> - Limit - <i>Landscaper will not place any additional terrain trees after the cap is reached.</i> - Replace - <i>Any terrain trees placed after the cap has been reached will instead be spawned as GameObjects. Note that spawning a large quantity of GameObjects can take a long time.</i>

Runtime Use

Landscaper *can* be used to generate foliage at runtime by calling the Foliage Area component's `Simulate()` function after having set all of the settings. Note that foliage simulation takes a little while to process (largely dependent on your settings) so there may be several seconds of delay before the foliage instances are placed. One other possible concern with runtime-generated foliage is that it won't benefit from static batching, resulting in more draw calls (and possibly worse performance) than would otherwise be the case if the same Biome was used to generate foliage in-editor.

Grass Placement Component (Experimental)

Landscaper also contains a component for planting small detail objects (such as grass) on the terrain at runtime. This is done in an area around the player to avoid having to spawn objects across the entire playable area.

The component can be found at 'Components > Landscaper > Experimental > Grass Placement'. Attach this to your player object and grass will be spawned in an area around them. **NOTE:** Grass is only planted on terrain objects.

Settings

Spacing

The distance between each grass instance. Lowering this number exponentially increases the number of grass instances spawned so it's best to keep it as high as possible for performance reasons.

Range

How far away (in Unity units) grass instances will be spawned. It's best to keep this as low as possible to avoid spawning too many objects.

Density

How dense the grass should be. A value lower than 1 will give grass a chance to not be placed, leaving a gap where it would have appeared.

Grass Types

Settings for each of the grass types that can be spawned. Pressing the 'Edit' button beside an entry in the list will open a separate window containing settings for the grass type. See [Grass Type Settings](#) below for more information.

Grid Size

Grass instances are updated in a grid, allowing instances to be created/destroyed in chunks. Larger grid sizes result in less frequent, but longer updates.

Jitter

How much to randomise the position of each instance. Since grass is placed in a grid, not jittering the position by a small amount will result in very uniform placement.

Stagger Updates

If checked, grid cell updates will happen over multiple frames. This is supposed to reduce frame hitching but can sometimes have the opposite effect.

Cell Update Rate

If 'Stagger Updates' is checked, this is the maximum number of grid cells that are allowed to be updated each frame.

Grass Type Settings

Prefab

Which prefab to spawn for each instance of this grass type.

Weight

How likely this grass type is to spawn (relative to every other grass type).

Random Rotation?

If checked, the grass will have a random rotation around its up direction.

Align to Surface?

If checked, the grass will rotate to match the underlying terrain geometry.

Scale

The minimum and maximum scale for each grass object. Scale is chosen at random from these values.

Slope Angle

The minimum and maximum slope angle this grass can grow on.

Layer Threshold

Any terrain layer in the mask below must have at least this much weight for the grass to be planted on it.

Terrain Layers

Which terrain layers the grass can grow on.